INCH-POUND
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DETAIL SPECIFICATION

PRINTED WIRING ASSEMBLY, SPECIAL CIRCUITS

This specification is approved for use by U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Special Circuits Printed Wiring Assembly referred to herein as the PWA. This is one of six printed wiring assemblies which comprise the Simplified Test Equipment/Internal Combustion Engine-Reprogrammable (STE/ICE-R) Vehicle Test Meter (VTM) (see Drawing 12259265).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5998

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines (see 4.2.1).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF DEFENSE

12258774 - Printed Wiring Assembly, Special Circuits.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/IPC J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies.

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(Application for copies should be addressed to American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample PWA shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Design, materials and manufacturing process selection shall be as specified herein and in applicable referenced specifications, standards, and drawings. Materials shall be uniform and free from imperfections or defects which affect their performance and serviceability. All metallic parts shall be made from corrosion resistant metals or treated with corrosion-resistant materials. Asbestos and Cadmium materials shall not be used in any form in any part of the PWA. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. Each PWA shall provide the following functional, operational, and performance capabilities.

3.3.1 Input power. The PWA shall require not more than the current specified in table I.

TABLE I. Input power.

Source	Voltage	Current
P1-5,6	5 Vdc 1/	0.030 A 2/
P1-13,14	15 Vdc	0.060 A
P1-19,20	-15 Vdc	0.060 A

1/ Vdc = Volts direct current.
2/ A = Ampere.

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3.3.2 Output voltages.

3.3.2.1 12 Volts (V) reference. The voltage at P1-9 and P1-10 shall be 12.0 ± 0.5 Vdc. U25 shall be capable of sourcing 100 milliamperes (mA) while maintaining the specified output voltage.

3.3.2.2 8 V reference. The voltage at P1-7 and P1-8 shall be 8.0 ± 0.4 Vdc. U24 shall be capable of sourcing 100 mA while maintaining the specified output voltage.

3.3.2.3 -8 V reference. The voltage at P1-17 and P1-18 shall be -8.0 ± 0.4 Vdc. U26 shall be capable of sourcing 200 mA while maintaining the specified output voltage.

3.3.3 Complementary metal oxide semiconductor (CMOS) logic levels. Unless otherwise specified (see 6.2), CMOS logic levels accepted by and output from the PWA shall be as follows:

Input High (Logic Level 1) = 3.5 Vdc minimum (min.)
Input Low (Logic Level 0) = 1.5 Vdc maximum (max.)
Output High (Logic Level 1) = 4.5 Vdc min.
Output Low (Logic Level 0) = 0.5 Vdc max.

3.3.4 Latches. (NOTE: A "/" following a capitalized logic name denotes logic negation.)

3.3.4.1 Output clocks. The signal at U27-4 shall be determined by the signals at P1-31 (IOB/3) and P1-37 (OEN-Q/) and shall conform to the waveforms shown in figure 1. The signal at U27-11 shall be determined by the signals at P1-32 (IOC/4) and P1-37 (OEN-Q/) and shall conform to the waveforms shown in figure 2. The signal at U27-10 shall be determined by the signals at P1-33 (IOD/5) and P1-37 (OEN-Q/) and shall conform to the waveforms shown in figure 3. The signal at U27-3 shall be determined by the signals at P1-34 (IOE/6) and P1-37 (OEN-Q/) and shall conform to the waveforms shown in figure 4.

3.3.4.2 Output latches. Logic levels at P1-21 through P1-26 (BUS 0 through BUS 5) shall be latched by U16 and appear at U16 pins 7, 5, 2, 10, 12, and 15, respectively, when clocked by U27-4 as shown in figure 1. Logic levels at P1-21 through P1-26 shall be latched by U17 and appear at U17 pins 7, 5, 2, 10,12, and 15, respectively, when clocked by U27-11 as shown in figure 2. Logic levels at P1-21 through P1-26 shall be latched by U18 and appear at U18 pins 7, 5, 2, 10, 12 and 15, respectively, when clocked by U27-10 as shown in figure 3, except a logic level 1 at U18-2 shall be 2.4 Vdc. Logic levels at P1-21 through P1-28 (BUS 0 through BUS 7) shall be latched by U19 and U20 and appear at U19 pins 7, 13, 14, and 4, and U20 pins 14, 4, 7, and 13, respectively, when clocked by U27-3 as shown in figure 4.

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3.3.5 Voltage level shifter. Logic levels present at U10 pins 3, 6, 10, and 14 shall be shifted as follows and shall be present at U10 pins 4, 5, 11, and 13, respectively, within 600 nanoseconds (ns) of their application to the input:

Output High (Logic Level 1) = [(voltages at U10 pin 16) - 0.30 Vdc] min.
Output Low (Logic Level 0) = 0.5 Vdc max.

3.3.6 Constant current source.

3.3.6.1 1.0 mA current. When the resistor network shown in figure 5 is connected to P1-98 and P1-100 and a logic level 1 is present at U18 pin 2 and U10 pins 6 and 10, the voltage across the 1 kilohm (kohm) resistor shall be adjustable to 1.000 $\pm$ 0.002 V.

3.3.6.2 0.1 mA current. When the resistor network shown in figure 6 is connected to P1-98 and P1-100 and a logic level 1 is present at U18 pin 2 and a logic level 0 at U10 pins 6 and 10, the voltage across the 10 kohm resistor shall be adjustable to 1.000 $\pm$ 0.002 V.

3.3.6.3 Current source off. With a logic level 0 present U18 pins 2 and U10 pins 6 and 10, there shall be no continuity between U15 pin 1 and pin 7 nor between U15 pin 14 and pin 8.

3.3.7 Digital to Analog Converter (DACON).

3.3.7.1 DACON min and max. When the logic levels representing 0 hexadecimal (HEXAD) are applied at U12 pins 13, 12, 11, 10, 9, 8, 7, 6, 5, and 4, the voltage at P1-55 shall be -3.750 $\pm$ 0.599 Vdc. When the logic levels representing 0400 HEXAD are applied at U12 pins 13, 12, 11, 10, 9, 8, 7, 6, 5, and 4, the voltage at P1-55 shall be +3.750 $\pm$ 0.599.

3.3.7.2 DACON conversion. When each of the HEXAD values in table II is applied at U12 pins 13, 12, 11, 10, 9, 8, 7, 6, 5, and 4, the voltage measured at P1-55 shall be that computed by the formula below with a tolerance of $\pm$ 0.015 Vdc. The min. and max. values shall be those measured in 3.3.7.1.

$$\text{output} = \frac{\text{(DACON max. - DACON min.)}}{\text{3FF HEXAD}} \times \text{(HEXAD input)} + \text{min.}$$

TABLE II. Hex inputs for DACON conversion.

0000	0022	0044	0066	0089	00AB	00CD	00EF
0111	0133	0155	0177	019A	01BC	01DE	0200
0222	0244	0266	0288	02AA	02CD	02EF	0311
0333	0355	0377	039A	03BC	03DE	03FF	

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3.3.8 Divide by n+1 counter. The frequency of the signal at U9 pins 14 and 15 shall be the frequency at U3 pin 7 divided by the sum of one and the decimal number represented by the logic levels at U9 pins 4, 5, 6, 7, 10, and 11.

3.3.9 Interrupt and flag select. When a logic level 0 is present at U21 pins 11, 10, and 9, the frequency of the signal at U21 pin 14 shall be the same as the frequency of the signal at U9 pins 14 and 15. The signal at U21 pin 14 shall be at P1-64 no greater than 250 ns later. When a logic level 0 is present at U21 pins 10 and 9 and a logic level 1 is present at U21 pin 11, the frequency of the signal at U21 pin 14 shall be the same as the frequency of the signal at U3 pin 7. The signal at U21 pin 14 shall be at P1-64 no greater than 250 ns later.

3.3.10 Test push button debouncer. With a logic level 0 at P1-50 and P1-52 not connected, U6 pin 3 and P1-56 shall be at logic level 0. When a logic level 0 is applied at P1-52 and P1-50 is open-circuited, U6 pin 3 and P1-56 shall change to a logic level 1 in no greater than 400 ns. When a logic level 0 is applied at P1-50 and P1-52 is open-circuited, U6 pin 3 and P1-56 shall change to a logic level 0 in no greater than 400 ns.

3.3.11 Microprocessor interrupt.

3.3.11.1 Microprocessor interrupt signal. With a logic level 1 present at U6 pin 5, a logic level 0 shall appear at P1-62 no greater than 1050 ns after the rising edge of the signal at U6 pin 3. With a logic level 1 present at U6 pin 9, a logic level 0 shall appear at P1-62 no greater than 1050 ns after the rising edge of the signal at U6 pin 11. A logic level 0 shall appear at P1-60 no greater than 300 ns after the rising edge of the signal at U6 pin 11.

3.3.11.2 Interrupt reset. When a logic level 1 is applied at P1-99, a logic level 1 shall appear at P1-62 no greater than 330 ns later.

3.3.12 Ignition signal pulse shaper. The frequency of the signal at U3 pin 6 shall be the same as the frequency of a signal applied at P1-86. The frequency can range from 2 Hertz (Hz) to 700 Hz. The rising edge of the signal at U3 pin 6 shall occur between 31.3 and 63.2 microseconds (µs) after the rising edge of the signal at P1-86. The positive pulse width of the signal at U3 pin 6 shall be 2.5 $\pm$ 0.5 milliseconds (ms).

3.4.13 Ignition kill signal. When U7 pins 2 and 5 are at logic level 1, the signal present at U9 pins 14 and 15 shall be observed identically at P1-48. When U7 pin 2 is at logic level 0, the signal at P1-48 shall be a logic level 1 and P1-70 shall be capable of sinking not less than 36 mA.

3.4.14 Pulse tachometer debouncer. When a 0 to 5 Vdc square wave of frequency between 5 and 325 Hz is applied at P1-66, the frequency of the signal at U3 pin 5 shall be twice the frequency of the applied signal and shall have a positive pulse width of 2.41 $\pm$ 1.08 ms. The

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rising edge of the signal at U3 pin 5 shall occur between 343 and 454 µs after the rising edge of the signal at P1-66.

3.3.15 Full wave rectifier.

3.3.15.1 Full wave rectifier on. With a logic level 1 at U13 pins 10 and 11 and a signal with amplitude of 2 V alternating current (ac) peak-to-peak at a frequency of 100 Hz applied at P1-78, the signal measured at P1-76 shall be positive. The signal at U13-4 shall have a positive pulse of twice the input. The signal at U13-4 shall have no more than 30 millivolts direct current (mVdc) difference between adjacent peaks and no more than 50 mVdc difference between adjacent low points.

3.3.15.2 Full wave rectifier off. With a logic level 0 at U13 pins 10 and 11 and a signal with amplitude between 100 mVdc and 5 Vdc and frequency between 10 and 500 Hz applied at P1-78, the signal at P1-76 shall have the same frequency as the signal applied at P1-78. With a signal source impedance at P1-78 of less than 100 ohms and an output impedance at P1-76 of greater than 50 kohms, the amplitude of the signal at P1-76 shall not be less than 95 percent (%) of the amplitude of the signal at P1-78.

3.3.16 Multiplexor. Logic levels present at U3 pins 6, 5, 4, or 3 shall be present at U3 pin 7 according to the logic levels at U3 pins 14 and 2 and table III. Logic levels shall be present at U3 pin 7 no greater than 490 ns after they are applied at U3 pins 6, 5, 4, and 3.

TABLE III. Multiplexor.

Input: U3 Pin 14	Input: U3 Pin 2	Output to U3 pin 7 from pin:
0	0	6
0	1	5
1	0	4
1	1	3

3.3.17 ATE Test. The PWA shall pass TACOM approved ATE test installed in TACOM approved ATE or equivalent.

3.3.18 Soldering. Soldering shall meet or exceed the requirements of ANSI/IPC J-STD-001.

3.4 Interface requirements. Each PWA shall accommodate the following requirements.

3.4.1 Overall envelope. Overall envelope of the PWA shall be in accordance with drawing 12258774.

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3.5 Ownership and support requirements.

3.5.1 Interchangeability. Interchangeability tolerances should permit parts, subassemblies and assemblies to be used in their parent assemblies without regard to the source of supply or manufacturer. Parts, subassemblies and assemblies having the full range of dimensions and characteristics permitted by the specification governing the part, subassembly or assembly should be usable as replacement items without selection and without departure from the specified performance guidelines of the parent items.

3.5.2 Identification marking. Unless otherwise specified (see 6.2), identification marking shall be permanent and legible and shall include as a minimum the following:

a. Nomenclature “PRINTED WIRING ASSEMBLY, SPECIAL CIRCUITS”.
b. Military part number.
c. Manufacturer’s CAGE code and name.
d. Manufacturer’s serial numbers.
e. Contract number.
f. Symbol “US”.

3.6 Operating environment requirements. Each PWA shall operate under the following environmental conditions without damage or loss of performance.

3.6.1 Temperature.

3.6.1.1 High temperature (operating). The PWA shall be capable of operation without damage or malfunction at a high temperature of +125°F (52°C) (without solar radiation).

3.6.1.2 Low temperature (operating). The PWA shall be capable of operation without damage or malfunction at a low temperature of +20°F (-7°C). For ambient temperatures below +20°F (-7°C), the test set shall be operated in a shelter.

3.6.1.3 High temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage at an ambient temperature of +160°F (71°C).

3.6.1.4 Low temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage in ambient temperatures of -60°F (-51°C).

3.6.2 Fungus. All nonmetallic materials that are not inherently fungistatic shall function in a fungus environment without adversely affecting the operation/performance of the PWA.

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3.6.3 Shock. The PWA shall evidence no degradation of function and meet the requirements of 3.3 when subjected to the shock test.

3.6.4 Vibration. The PWA shall evidence no degradation of function and meet the requirements of 3.3 when subjected to the vibration test.

3.6.5 Burn-in. Unless otherwise specified (see 6.2), the PWA shall be subjected to and pass a "power on" burn-in prior to the inspections listed in 4.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspections (CI) (see 4.1.2).

4.1.1 First article inspection. Unless otherwise specified (see 6.2 and 6.3), the Government shall select one PWA produced under the production contract for first article inspection. First article inspection shall include all the verifications listed in table IV.

TABLE IV. Verification methods.

Title	Requirement	Verification
Design, materials, & manufacturing processes	3.2	4.2
Operating requirements:	3.3	4.2.3
Input power	3.3.1	4.2.3.1
Output voltages	3.3.2	4.2.3.2
Logic levels	3.3.3	4.2.3.3
Latches	3.3.4	4.2.3.4
Voltage level shifter	3.3.5	4.2.3.5
Constant current source	3.3.6	4.2.3.6
DACON	3.3.7	4.2.3.7
Divide by n+1 counter	3.3.8	4.2.3.8
Interrupt and flag select	3.3.9	4.2.3.9
Test push button debouncer	3.3.10	4.2.3.10
Microprocessor interrupt signal	3.3.11	4.2.3.11
Ignition signal pulse shaper	3.3.12	4.2.3.12
Ignition kill signal	3.3.13	4.2.3.13
Pulse tachometer debouncer	3.3.14	4.2.3.14
Full wave rectifier	3.3.15	4.2.3.15
Multiplexor	3.3.16	4.2.3.16

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TABLE IV. Verification method - Continued.

Title	Requirement	Verification
ATE Test 1/	3.3.17	4.2.3.17
Interface requirements:	3.4	4.2.4
Overall envelope & interfaces	3.4.1	4.2.4.1
Ownership & support requirements	3.5	4.2.5
Interchangeability	3.5.1	4.2.5.1
Identification marking	3.5.2	4.2.5.2
Operating environment requirements	3.6	4.2.6
Temperature	3.6.1	4.2.6.1
Fungus	3.6.2	4.2.6.2
Shock	3.6.3	4.2.6.3
Vibration	3.6.4	4.2.6.4

1/ Not required unless specified (see 6.2).

4.1.2 Conformance Inspection (CI) 100% inspection. CI shall be conducted on all items (100% inspection) and shall include those examinations and tests from table IV as specified in the contract.

4.2 Verification methods. Acceptable verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

4.2.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following standard (room) ambient conditions:

a. Temperature 77 ± 18°F.
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

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4.2.3 Operating requirements verification.

4.2.3.1 Input power. To determine conformance to 3.3.1, the input current shall be measured with an ammeter.

4.2.3.2 Output voltages.

4.2.3.2.1 12 V reference. To determine conformance to 3.3.2.1, the voltage at P1-9 and P1-10 with respect to P1-1 shall be measured with a voltmeter. A 110 ohm, 5%, 2 watt resistor shall be put at P1-9 and the current from the unterminated end of the resistor of P1-1 shall be measured with an ammeter.

4.2.3.2.2 8 V reference. To determine conformance to 3.3.2.2, the voltage at P1-7 and P1-8 with respect to P1-1 shall be measured with a voltmeter. A 70 ohm, 5%, 2 watt resistor shall be put at P1-7 and the current from the unterminated end of the resistor to P1-1 shall be measured with an ammeter.

4.2.3.2.3 -8 V reference. To determine conformance to 3.3.2.3, the voltage at P1-17 and P1-18 with respect to P1-1 shall be measured with a voltmeter. A 35 ohm, 5%, 2 watt resistor shall be put at P1-17 and the current from the unterminated end of the resistor to P1-1 shall be measured with an ammeter.

4.2.3.3 CMOS logic levels. To determine conformance to 3.3.3, the input and output CMOS logic levels shall be observed and verified with an oscilloscope.

4.2.3.4 Latches.

4.2.3.4.1 Output clocks. To determine conformance to 3.3.4.1, a logic analyzer shall be used to verify that the signal at U27-4 conforms to the waveforms shown in figure 1, the signal at U27-11 conforms to the waveforms shown in figure 2, the signal at U27-10 conforms to the waveforms shown in figure 3, and the signal at U27-3 conforms to the waveforms shown in figure 4.

4.2.3.4.2 Output latches. To determine conformance to 3.3.4.2, a logic analyzer shall be used to verify that the logic levels applied to BUS 0 through BUS 7 (P1-21 through P1-28) appear at the outputs specified in 3.3.4.2 when signals conforming to those shown in figures 1, 2, 3, and 4 are applied to the PWA.

4.2.3.5 Voltage level shifter. To determine conformance to 3.3.5, a logic level 1, shall be applied at P1-38. A hexadecimal pattern 08 shall be applied at P1-21 through P1-28 and the waveform of figure 2 shall be applied. A HEXAD pattern 18 shall be applied at P1-21 through

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P1-28 and the waveform of figure 3 shall be applied. The voltage levels at U10 pins 13, 11, 5 and 4 shall be measured with a voltmeter.

4.2.3.6 Constant current source.

4.2.3.6.1 1.0 milliamperes (mA) current. To determine conformance to 3.3.6.1, a HEXAD pattern 04 shall be applied at P1-21 through P1-28 and the waveform of figure 3 shall be applied. Connect the resistor network shown in figure 5 at P1-98 and P1-100. Adjust R25 until the voltage across the 1 kohm resistor is 1.000 $\pm$ 0.002 V.

4.2.3.6.2 0.1 mA current. To determine conformance to 3.3.6.2, a HEXAD pattern OC shall be applied at P1-21 through P1-28 and the waveform of figure 3 shall be applied. Connect the resistor network shown in figure 6 at P1-98 and P1-100. Adjust R28 until the voltage across the 10 kohm resistor is 1.000 $\pm$ 0.002 V.

4.2.3.6.3 Current source off. To determine conformance to 3.3.6.3, a HEXAD pattern 00 shall be applied at P1-21 through P1-28 and the waveform of figure 3 shall be applied. Connect a 1 kohm resistor (exact value not important) from U15 pin 1 to pin 7. Verify that the resistance between U15 pin 1 and pin 7 has the same resistance value as the resistor connected between the pins. Connect the resistor from U15 pin 8 to pin 14. Verify that the resistance between U15 pin 8 and pin 14 has the same resistance value as the resistor connected between the pins.

4.2.3.7 DACON.

4.2.3.7.1 DACON min and max. To determine conformance to 3.3.7.1, a HEXAD 00 shall be applied at P1-21 through P1-28 and the waveform of figure 3 shall be applied. The waveform of figure 4 shall then be applied. The voltage at P1-55 shall be measured with a voltmeter, verified, and recorded for use in 4.2.3.7.2. A HEXAD 04 shall be applied at P1-21 through P1-28 and the waveform of figure 3 shall be applied. A HEXAD 00 shall then be applied to P1-21 through P1-28 and the waveform of figure 4 shall be applied. The voltage at P1-55 shall be measured with a voltmeter, verified, and recorded for use in 4.2.3.7.2.

4.2.3.7.2 DACON conversion. For each HEXAD input given in table II, the first two digits of the HEXAD pattern shall be applied at P1-21 through P1-28 and the waveform of figure 3 shall be applied. The last two digits of the HEXAD pattern and the waveform of figure 4 shall be applied. The voltage at P1-55 shall be measured with a voltmeter. The theoretical voltage shall be computed using the formula in 3.3.7.2, the values measured in 4.2.3.7.1, and the appropriate HEXAD value from table II. Verify that the measured output conforms to the computed value within the tolerance specified in 3.3.7.2.

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4.2.3.8 Divide by n + 1 counter. A 0 to 5 Vdc, 700 kilohertz (kHz) square wave shall be applied at P1-51. A HEXAD pattern 02 shall be applied at P1-21 through P1-28 and the waveform of figure 2 shall be applied. For each HEXAD input specified in table V, the HEXAD pattern shall be applied at P1-21 through P1-28 and the waveform of figure 1 shall be applied. The frequency of the signal at U9 pins 14 and 15 shall be measured. This measured frequency shall be verified to conform to the corresponding limits specified in table V for the appropriate HEXAD input to determine conformance to 3.3.8.

TABLE V. Divide by n + 1 counter. ($n \neq 0$).

Hex Input	Frequency Output
04	140.0 kHz
09	70.0 kHz
0E	46.7 kHz
13	35.0 kHz
18	28.0 kHz
1D	23.3 kHz
22	20.0 kHz
27	17.5 kHz
2C	15.6 kHz
31	14.0 kHz
36	12.7 kHz
3B	11.7 kHz
40	10.8 kHz

4.2.3.9 Interrupt and flag select. To determine conformance to 3.3.9, a 0 to 5 Vdc, 700 kHz square wave shall be applied to P1-51. A HEXAD pattern 03 shall be applied at P1-21 through P1-28 and the waveform of figure 2 shall be applied. A HEXAD pattern 01 shall be applied at P1-21 through P1-28 and the waveform of figure 1 shall be applied. The frequency of the signal at U3-7 shall be verified to be 700 kHz. A HEXAD pattern 0A shall be applied at P1-21 through P1-28 and the waveform of figure 2 shall be applied. The frequency of the signal at U21 pin 14 shall be verified to be 350 kHz.

4.2.3.10 Test push button debouncer. To determine conformance to 3.3.10, a logic level 0 shall be applied at P1-50 and U6 pin 3 shall be verified to be at logic level 0. Simultaneously a logic level 0 shall be applied to P1-52, and P1-50 shall be allowed to float. U6 pin 3 shall be verified to change to a logic level 1 in less than 400 ns. Simultaneously a logic level 0 shall be applied to P1-50, and P1-52 shall be allowed to float. U6 pin 3 shall be verified to change to a logic level 0 in less than 400 ns.

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4.2.3.11 Microprocessor interrupt.

4.2.3.11.1 Microprocessor interrupt signal. To determine conformance to 3.3.11.1, a HEXAD pattern 2A shall be applied at P1-21 through P1-28 and the waveform of figure 2 shall be applied. A 0 to 5 Vdc rising pulse edge shall be applied at P1-51 and the signals at U6 pins 11 and 12, P1-60, and P1-62 shall be observed.

4.2.3.11.2 Interrupt reset. To determine conformance to 3.3.11.2, a logic level 1 shall be applied at P1-99, and P1-62 shall be observed with an oscilloscope.

4.2.3.12 Ignition signal pulse shaper. To determine conformance to 3.3.12, a 0 to 8 V, 10 Hz square wave shall be applied at P1-86. The signals at U3 pin 6 and P1-86 shall be observed with an oscilloscope.

4.2.3.13 Ignition kill signal. To determine conformance to 3.3.13, a HEXAD pattern 04 shall be applied at P1-21 through P1-28 and the waveform of figure 2 shall be applied. A HEXAD pattern 00 shall be applied at P1-21 through P1-28 and the waveform of figure 1 shall be applied. A 150 ohm, 5%, 0.25 watt resistor shall be attached from P1-70 to a 5 Vdc source. A 5 V peak-to-peak, 10 Hz square wave at P1-86 and a logic level 1 shall be verified to be at P1-48. The current through the 150 ohm resistor shall be verified with an ammeter to conform to 3.3.13. A HEXAD pattern 20 shall be applied at P1-21 through P1-28 and the waveform of figure 3 shall be applied. The signal at P1-48 shall be observed.

4.2.3.14 Pulse tachometer debouncer. To determine conformance to 3.3.14, a 0 to 5 Vdc, 325 Hz square wave shall be applied at P1-66. The signal at U3 pin 5 shall be observed.

4.2.3.15 Full wave rectifier.

4.2.3.15.1 Full wave rectifier on. To determine conformance to 3.3.15.1, a HEXAD pattern 10 shall be applied at P1-21 through P1-28 and the waveform of figure 3 shall be applied. A 5 V peak-to-peak, 500 Hz sine wave shall be applied at P1-78 and the signals at P1-76 and U3 pin 4 shall be observed.

4.2.3.15.2 Full wave rectifier off. To determine conformance to 3.3.15.2, a HEXAD pattern 00 shall be applied at P1-21 through P1-28 and the waveform of figure 3 shall be applied. A 5 V peak-to-peak, 500 Hz sine wave shall be applied at P1-78 and the signals at P1-76 and U3 pin 4 shall be observed.

4.2.3.16 Multiplexor. To determine conformance to 3.3.16, the following procedure shall be performed:

a. Apply a 100 Hz 0 to 5 V square wave to P1-51.

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b. Apply a HEXAD pattern 03 to P1-21 through P1-28, and apply the waveform of figure 2.
c. An oscilloscope shall be used to verify that the square wave is present at U3-7, delayed by no more than 490 ns.
d. Apply the waveform of 4.2.3.12 to P1-86.
e. Apply a HEXAD pattern 00 to P1-21 through P1-28, and apply the waveform of figure 2.
f. An oscilloscope shall be used to verify that the signal at U3-6 is present at U3-7, delayed by no more than 490 ns.
g. Apply the waveform of 4.2.3.14 to P1-66.
h. Apply a HEXAD pattern 01 to P1-21 through P1-28, and apply the waveform of figure 2.
i. An oscilloscope shall be used to verify that the signal at U3-5 is present at U3-7, delayed by no more than 490 ns.
j. Apply the waveform of 4.2.3.15.1 to P1-78.
k. Apply a HEXAD pattern 02 to P1-21 through P1-28, and apply the waveform of figure 2.
l. An oscilloscope shall be used to verify that the signal at U3-4 is present at U3-7, delayed by no more than 490 ns.

4.2.3.17 ATE test. To determine conformance to 3.3.17, install the PWA into TACOM approved fixture (such as BTF-2500-03), on TACOM approved, ATE equipment (such as HP 3062) with TACOM approved software and instructions (such as BSW-25000030).

4.2.4 Interface requirements verification.

4.2.4.1 Overall envelope inspection. Insert the PWA into the present existing location of the test set, verify its envelope and interfacing conformance.

4.2.5 Ownership and support requirements verification.

4.2.5.1 Interchangeability inspection. To determine conformance to 3.5.1, the interchangeability between different parts, subassemblies, and assemblies having the same military part number shall be verified.

4.2.5.2 Identification marking inspection. Verify the presence of the required PWA identification marking.

4.2.6 Operating environment requirements verification. The electrical performance tests specified in 4.2.3 through 4.2.3.17 shall be conducted prior to and at the completion of each environmental test. Verification of performance during the high temperature and low temperature

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environmental tests shall be demonstrated by the tests specified in 4.2.3.1, 4.2.3.5, 4.2.3.6, 4.2.3.7, and 4.2.3.16.

4.2.6.1 Temperature. To determine conformance to 3.6.1, the PWA shall be tested as follows:

4.2.6.1.1 High temperature (operating). To determine conformance to 3.6.1.1, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure II (hot category), of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3.

4.2.6.1.2 Low temperature (operating). To determine conformance to 3.6.1.2, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure II, of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3.

4.2.6.1.3 High temperature (nonoperating). To determine conformance to 3.6.1.3, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure I, of MIL-STD-810, or equivalent (see 4.2.1).

4.2.6.1.4 Low temperature (nonoperating). To determine conformance to 3.6.1.4, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure I, of MIL-STD-810, or equivalent (see 4.2.1).

4.2.6.2 Fungus. To determine conformance to 3.6.2, the PWA shall be subjected to the applicable fungus test in accordance with method 508.4 of MIL-STD-810 (28 day test), or equivalent (see 4.2.1).

4.2.6.3 Shock. To determine conformance to 3.6.3. The PWA shall be subjected to the shock test in accordance with method 516.4, Procedure I, 40 g peak of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3 after shock test.

4.2.6.4 Vibration. To determine conformance to 3.6.4, the PWA shall be subjected to a vibration test using a composite of figures 7 through 9. These figures represent the cargo environment at the cargo bed of a composite of tactical wheeled vehicles, the 5-ton 813 truck and the 12-ton M127 semitrailer. The data include differing vehicle loading conditions traversing over specially designed courses ranging from paved highway to off road conditions at various vehicle speeds. The spectrum is broadband random with peaks and notches at various discrete frequency bands. Break points are provided for establishing the spectrum shape. The test duration is 500 miles. The PWA shall be tested to 3.3 after the vibration test.

4.2.6.5 Burn-in. When required (see 6.2), the burn-in shall consist of three temperature cycles and a six gravity unit (g) root mean square (rms) random vibration as follows:

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a. Temperature cycle. Each temperature cycle shall consist of 30 minutes at minus (-) 60 degrees Fahrenheit (°F) (-51 degrees Celsius (°C)) and 30 minutes at plus (+) 125°F (52°C). The rate of temperature change between temperature extremes shall be not less than 5°F (2.8°C) per minute. Power shall be applied to the PWA for not less than 30 minutes from 20°F (7°C) through the completion of each high-temperature soak.
b. Random vibration. Following all temperature cycling, the PWA shall be hard mounted to the vibration table so that the axis of vibration is perpendicular to the plane of the PWA. The vibration shall be at room ambient temperature, for a period of 10 minutes, as shown in figure 10.

NOTE: PWAs being procured for simultaneous installation in VTMs being procured do not have to be subjected to "burn-in" provided the VTMs are subjected to the 16-hour "power on" burn-in.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The PWA has 12, 8, and -8 Vdc voltage regulators (U25, U24, U26) for this and other printed wiring assemblies in the STE/ICE-R VTM. The PWA latches data from the Data Bus (P1-21 through P1-28) into U16, U17, U18, or U19 and U20 depending on the I/O control line used. Latched data is used to select or deselect the circuits to perform the following functions: A Constant Current Source (U15, U21, U22, U23) can source either 1.0 or 0.1 mA. A Digital-to-Analog Converter (DACON; U11, U12) provides -3.75 to 3.75 Vdc according to a 10-bit binary input. A four input digital Multiplexor (U3) can be controlled to route signals through a Divide by n+1 Counter (U9), which can selectively divide the input frequency by any number from 1 to 64. A Test Push Button Debouncer (U6, U7) provides a clean signal for the microprocessor interrupt signal and an event flag. The output of the Multiplexor or the Divide by

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n+1 Counter can be used as microprocessor interrupts and event flags as well. The Ignition Signal Pulse Shaper (U1, U4, U8) improves the quality of the ignition signal so that it can be measured using the Divide by n+1 Counter or so that it can selectively drive an interrupt signal. The Pulse Tachometer Deglitcher (U2, U4, U5) provides a clean, square wave signal of double the input frequency. The Full Wave Rectifier (U8, U13, U14) is software selectable. The PWA is military unique because it is designed specifically to be a component of the Simplified Test Equipment/Internal Combustion Engine - Reprogrammable (STE/ICE-R) Vehicle Test Meter (VTM) and to withstand environmental conditions which exceed commercial requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article inspection is required (see 3.1).
d. PWA drawing number, revision letter, date and PWA part number (see 3.2 and 3.5.2).
e. If CMOS logic levels should be other than as specified (see 3.3.3).
f. If marking is other than as specified (see 3.5.2).
g. If burn-in is not required (see 3.6.5).
h. If ATE test is required (see table IV footnote).
i. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table IV and their designated frequency based on a risk assessment for the procurement.

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6.5 Historical data. Historically, ATPD 2146, dated 9 March 1988, was superseded by MIL-P-62740(AT).

6.6 Subject term (key word) listing.

Automatic Testing Equipment (ATE) test
Ignition inhibits
Ignition signal outputs
Simplified Test Equipment/Internal Combustion Engine - Reprogrammable (STE/ICE-R)
Vehicle Test Meter (VTM)

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

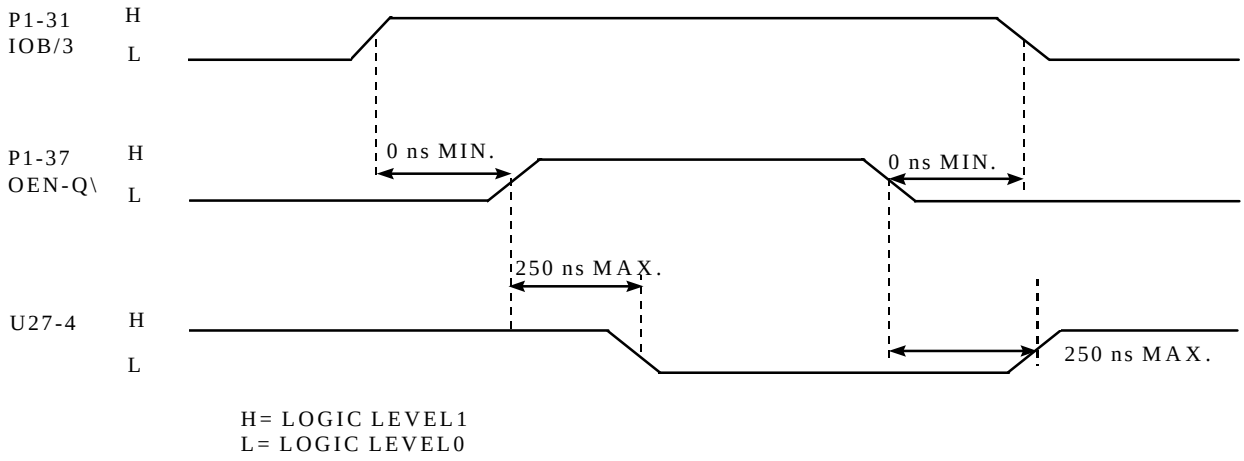


FIGURE 1. Waveform to generate U27-4 clock.

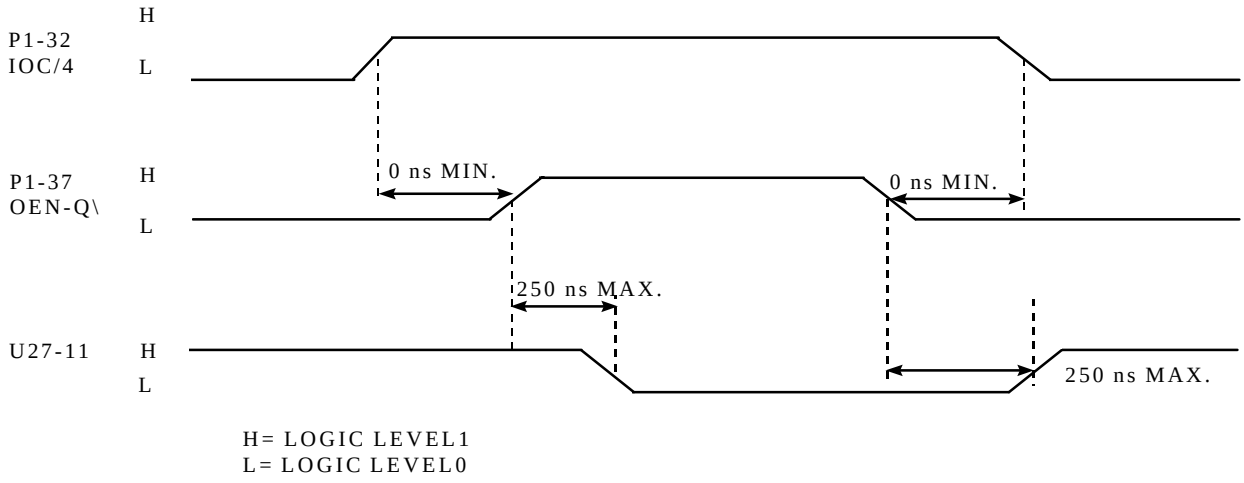
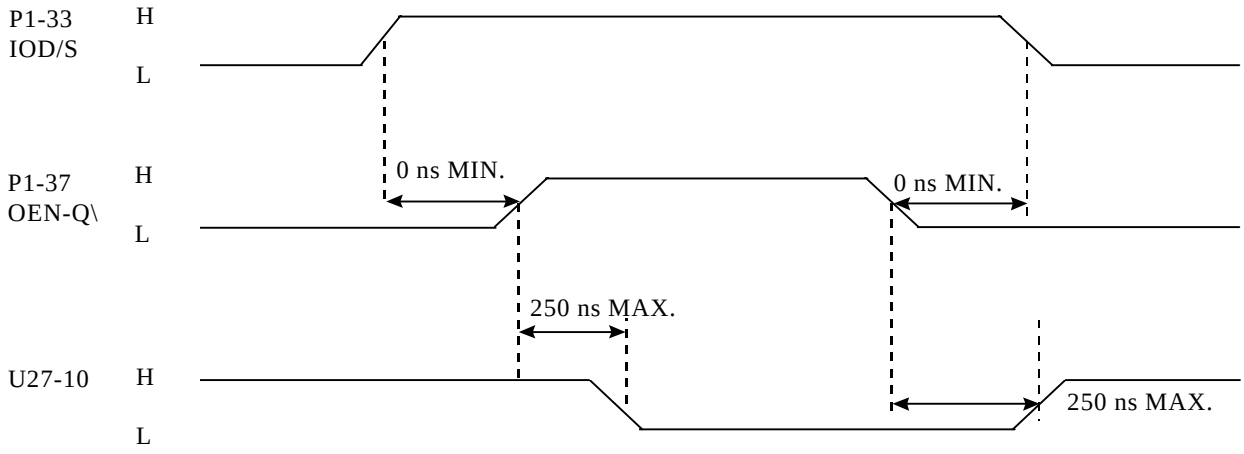


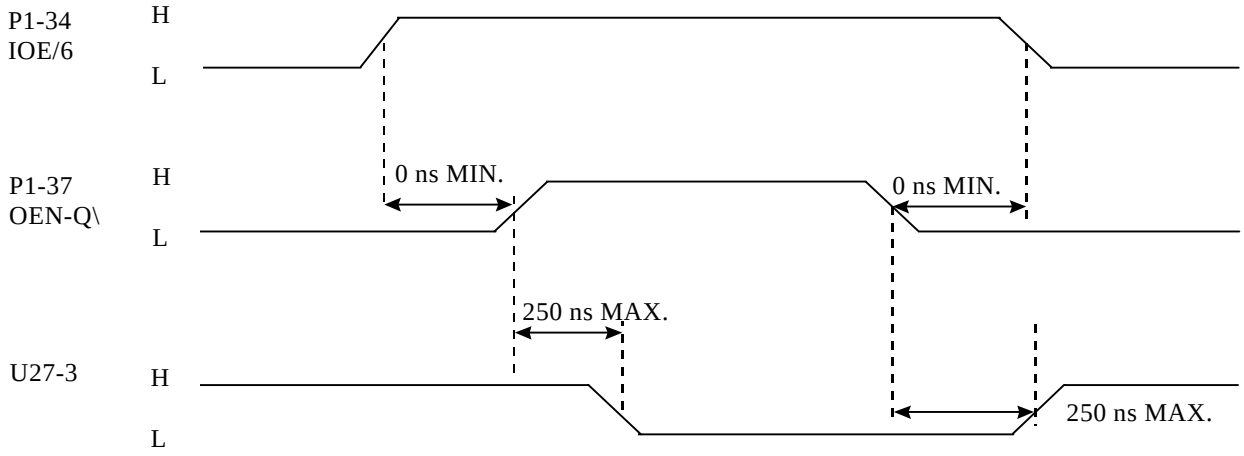
FIGURE 2. Waveforms to generate U27-11 clock.

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H= LOGIC LEVEL1
L= LOGIC LEVEL0

FIGURE 3. Waveforms to generate U27-10 clock.



H= LOGIC LEVEL1
L= LOGIC LEVEL0

FIGURE 4. Waveforms to generate U27-3 clock.

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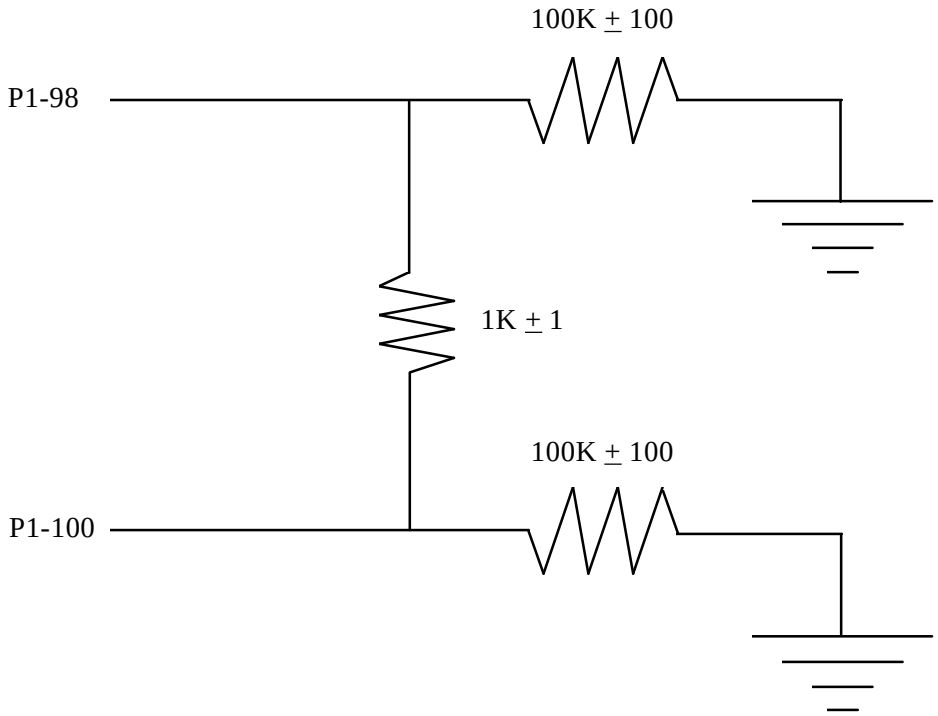


FIGURE 5. Resistor network for 1.0 mA current source testing.

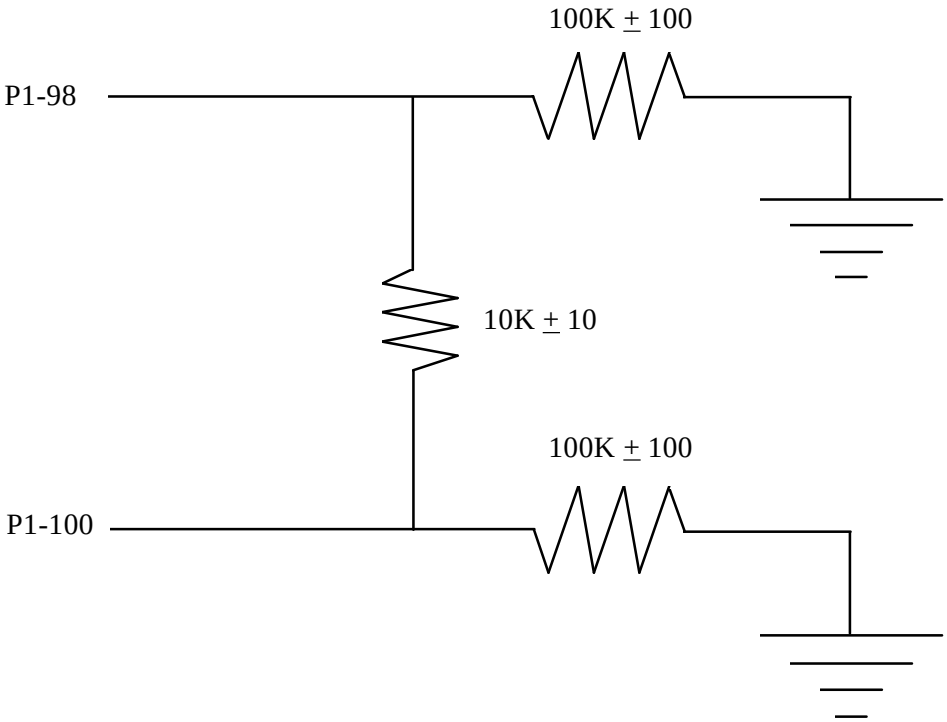
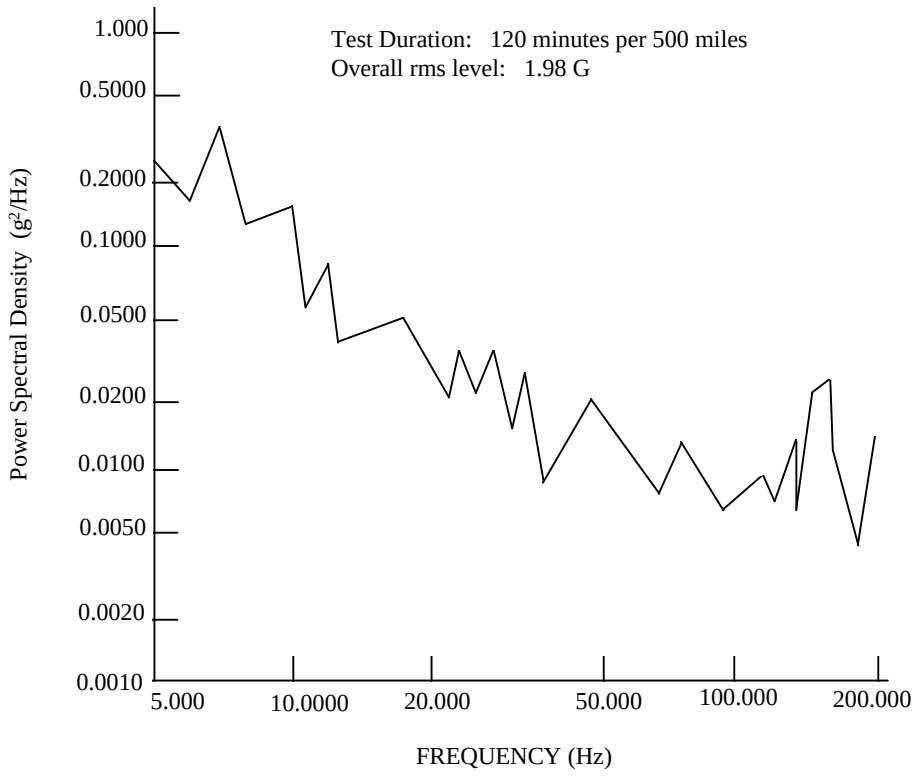


FIGURE 6. Resistor network for 0.1 mA current source testing.

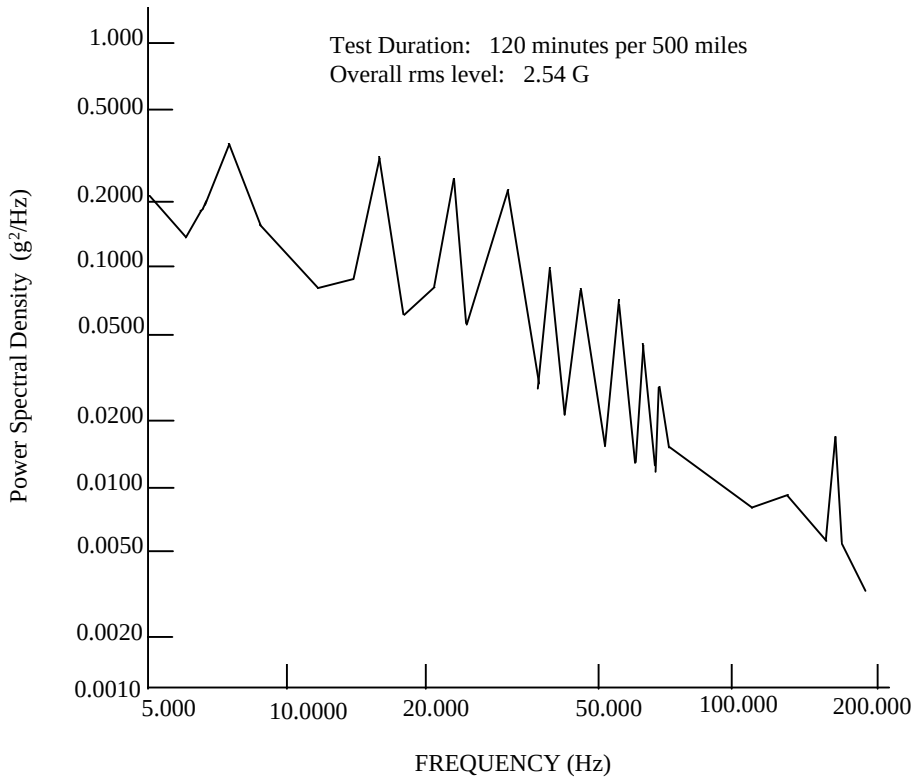
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BREAKPOINTS	
FREQ	PSD VALUE
5	0.23840
6	0.16840
7	0.38230
8	0.14870
10	0.16840
11	0.05930
12	0.09750
13	0.04090
18	0.05370
23	0.01940
24	0.03520
26	0.02310
28	0.03520
32	0.01670
33	0.03110
37	0.00810
49	0.02190
63	0.00700
73	0.01150
98	0.00660
120	0.00970
123	0.00680
138	0.01340
140	0.00760
146	0.02490
159	0.02820
163	0.01300
188	0.00480
200	0.01340

FIGURE 7. Basic transportation, composite tactical wheeled environment, vertical axis.

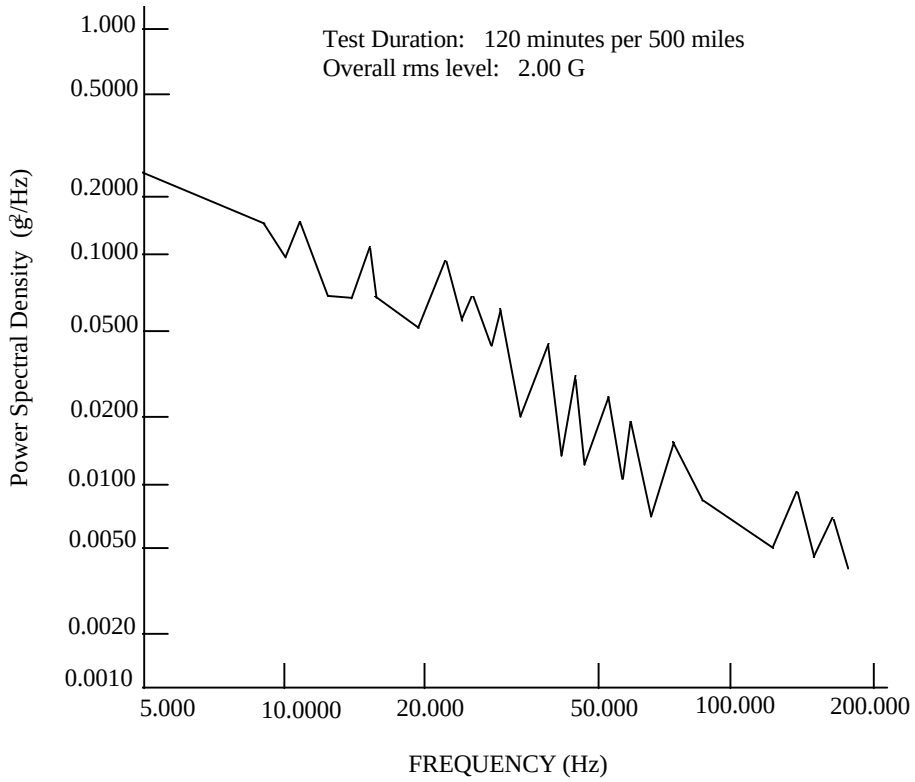
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BREAKPOINTS	
FREQ	PSD VALUE
5	0.22130
6	0.13140
7	0.20540
8	0.38230
9	0.15630
12	0.07990
15	0.08830
16	0.32940
19	0.06080
22	0.09280
24	0.26340
27	0.05640
32	0.22690
36	0.02820
40	0.11600
44	0.02090
48	0.08190
53	0.01480
56	0.07420
60	0.01270
64	0.04980
65	0.01100
68	0.03260
69	0.01480
114	0.00740
132	0.00830
160	0.00510
169	0.01710
173	0.00550
200	0.00310

FIGURE 8. Basic transportation, composite tactical wheeled environment, longitudinal axis.

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BREAKPOINTS	
FREQ	PSD VALUE
5	0.25490
10	0.13030
11	0.09670
12	0.13360
13	0.06500
15	0.06340
16	0.11230
17	0.06500
22	0.04590
24	0.08980
26	0.05070
28	0.06830
30	0.03670
32	0.05880
38	0.01920
40	0.03860
44	0.01290
48	0.02940
50	0.01140
56	0.02460
60	0.00960
63	0.01880
70	0.00660
78	0.01500
93	0.00730
137	0.00460
145	0.00910
162	0.00430
176	0.00660
200	0.00360

FIGURE 9. Basic transportation, composite tactical wheeled environment, transverse axis.

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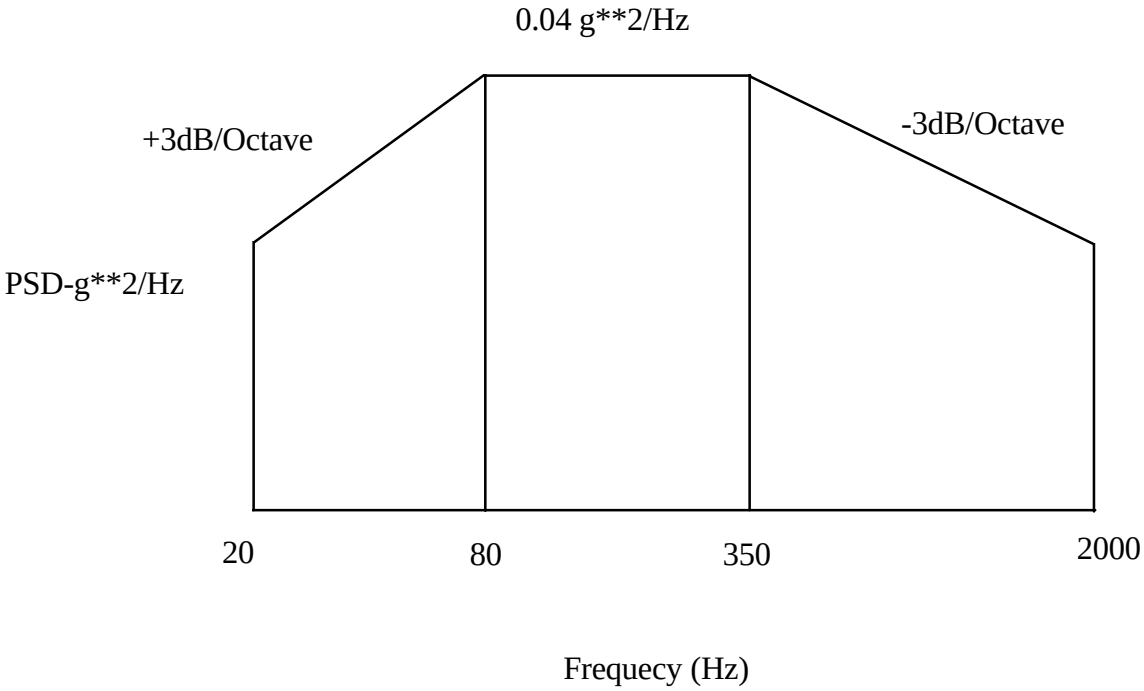


FIGURE 10. Random Vibration Spectrum.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 5998-0075)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-62740A(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 980123

3. DOCUMENT TITLE
Printed Wiring Assembly, Special Circuits

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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